AI Customer Feedback Intelligence System

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Problem Statement & Objectives

GOAL

- Automatically understand customer sentiment and classify reviews as Positive / Negative / Neutral
- Combine Supervised Learning (sentiment classification) with Unsupervised Learning (K-Means) to uncover the root causes behind negative reviews

DATASET

- Women's E-Commerce Clothing Reviews — 23,486 raw reviews
- 22,622 reviews after cleaning (missing values & duplicates removed)
- Two classifiers compared: Logistic Regression vs Random Forest
- K-Means clustering applied to the 2,369 negative reviews

23,486

Raw reviews collected

22,622

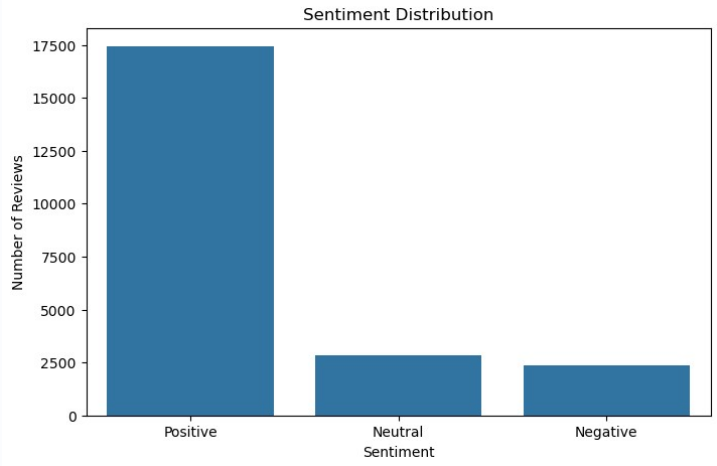
Reviews after cleaning

2,369

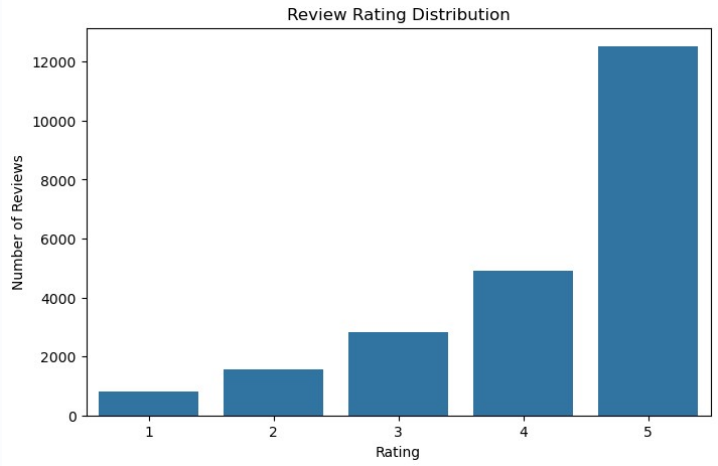
Negative reviews clustered

Exploratory Data Analysis

SENTIMENT DISTRIBUTION



REVIEW RATING DISTRIBUTION



KEY EDA NUMBERS

Metric	Value
Average rating	4.18
Positive reviews	17,430
Neutral reviews	2,823
Negative reviews	2,369

Data Cleaning & Text Preprocessing

DATA CLEANING

- 23,486 → 22,628 rows (dropna)
- 22,628 → 22,622 rows (drop_duplicates)
- Kept: Review Text, Rating, Class Name, Recommended IND
- Columns renamed for clarity (Product Category, Recommendation Indicator)

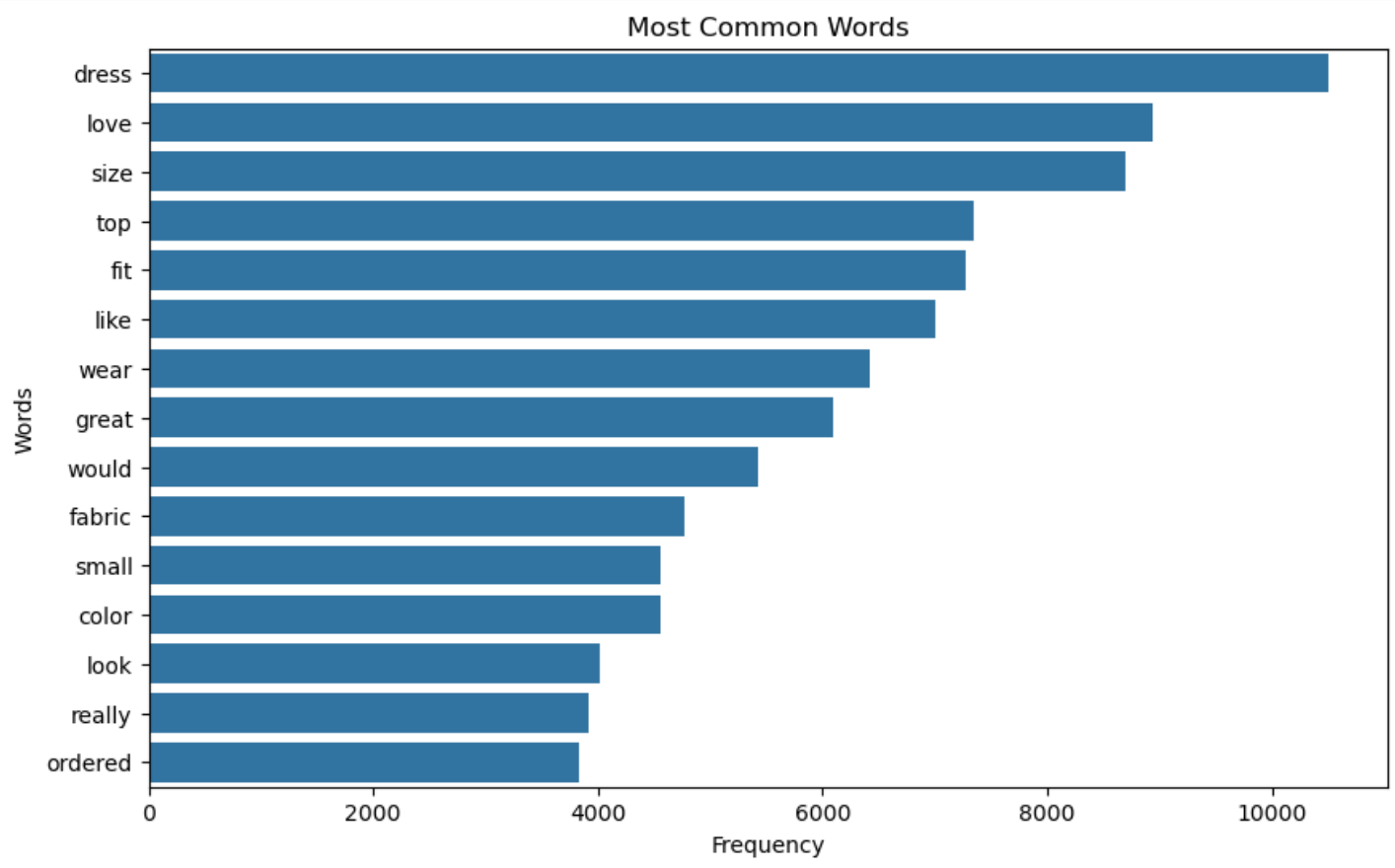
TEXT NORMALIZATION

- Lower-case the full review text
- Tokenize with NLTK word_tokenize
- Keep alphabetic tokens only (isalpha)
- Remove English stopwords (NLTK)
- Rejoin tokens → 'Clean Review' column

WHY THESE STEPS (TECHNICAL RATIONALE)

- isalpha() strips digits, punctuation and emoji — noise that TF-IDF cannot weight meaningfully
- Lower-casing prevents 'Great' and 'great' from being counted as two different features
- Stopword removal shrinks the vocabulary so TF-IDF focuses weight on content-bearing words
- dropna / drop_duplicates guarantee one clean record per genuine review

Most Common Words



WHY THIS MATTERS

- Confirms the cleaning pipeline worked — top words should be meaningful product/fit/quality terms, not stopwords or junk tokens
- Directly informs which words dominate the 5,000 TF-IDF features

Feature Extraction — TF-IDF

- TfidfVectorizer(max_features=5000) turns cleaned review text into numeric features
- Fitted on the training reviews only, then applied to the test reviews (no test-set leakage)
- Sentiment label derived from Rating: ≤ 2 is Negative = 3 is Neutral ≥ 4 is Positive
- Split 80/20 before vectorizing: train_test_split(stratify = y, random_state = 42) → 18,097 train / 4,525 test

Stage	Shape
Cleaned reviews (all)	22,622 texts
Training set (X_train)	18,097 × 5,000
Test set (X_test)	4,525 × 5,000

WHY TF-IDF & WHY STRATIFY

- TF-IDF down-weights common words and up-weights distinctive ones — better signal than raw word counts (Bag-of-Words)
- max_features=5000 caps dimensionality, controlling overfitting and memory use
- stratify=y keeps the 77/12/11 class ratio identical in train and test — critical given the imbalance
- random_state=42 makes the split reproducible for grading
- Split first, then fit TF-IDF on the training set only: test-set vocabulary and IDF weights must not leak into training

Classification Models & Results

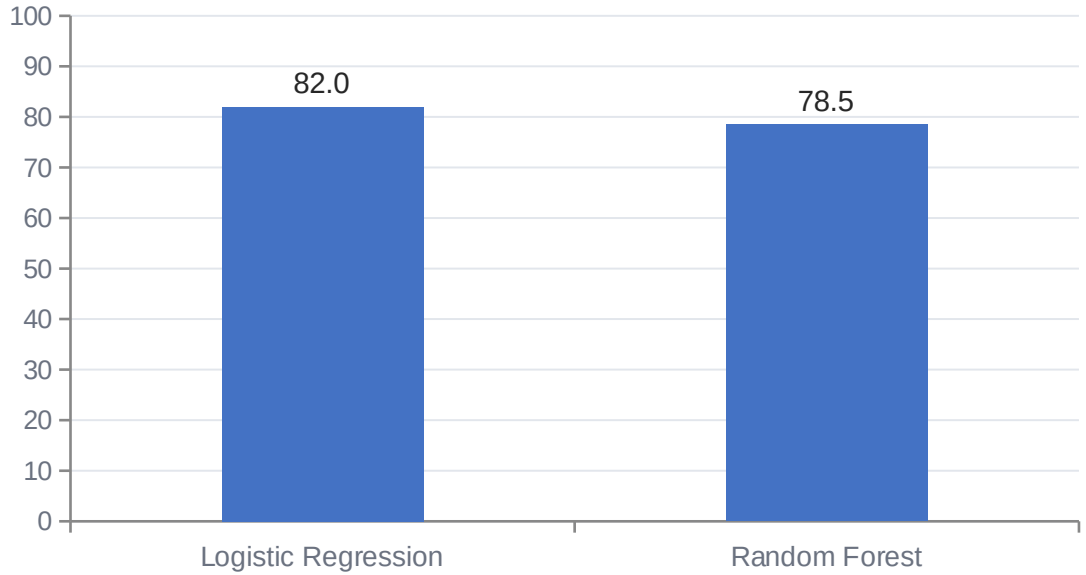
LOGISTIC REGRESSION — 82.0% ACCURACY

Class	Prec.	Recall	F1
Negative	0.61	0.41	0.49
Neutral	0.44	0.23	0.30
Positive	0.87	0.97	0.92

RANDOM FOREST — 78.5% ACCURACY

Class	Prec.	Recall	F1
Negative	0.64	0.14	0.23
Neutral	0.44	0.02	0.05
Positive	0.79	1.00	0.88

ACCURACY COMPARISON

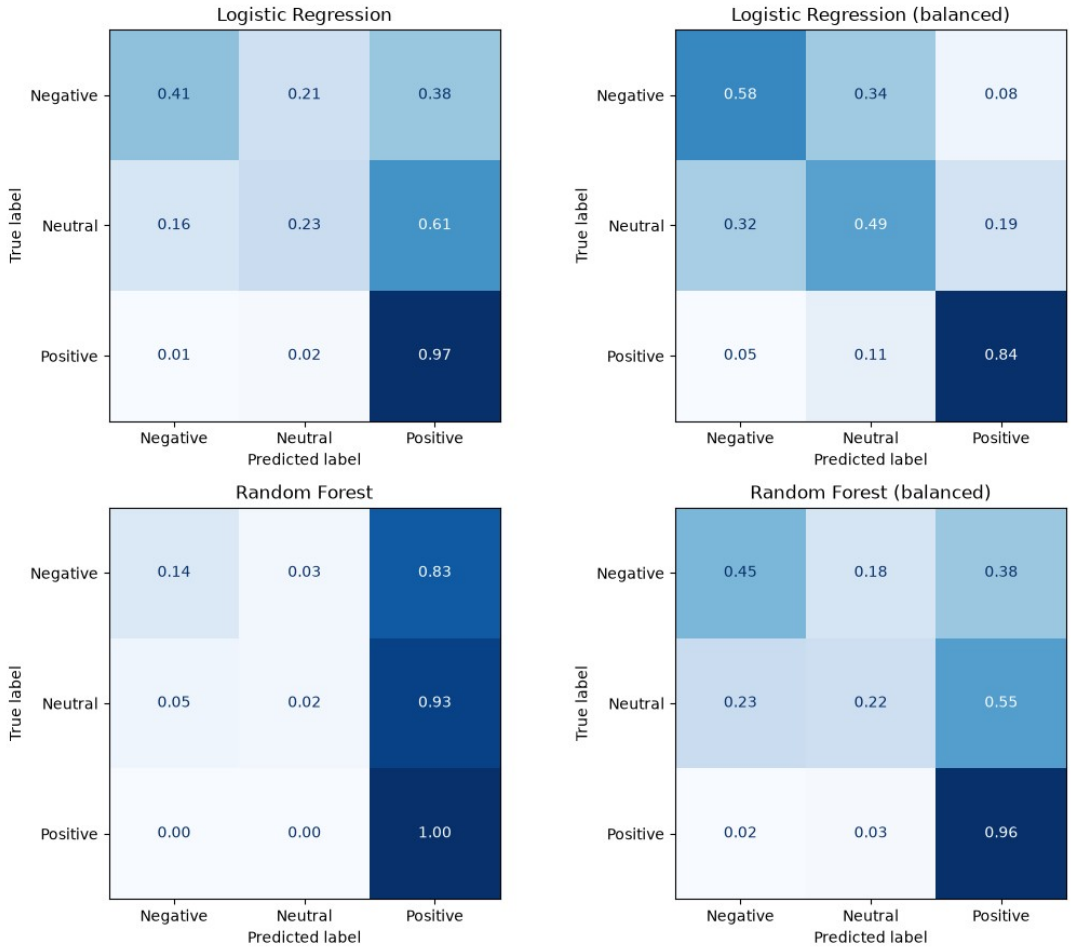


Conclusions

- Logistic Regression wins overall and, crucially, recovers far more Negative/Neutral reviews
- Random Forest collapses onto the majority Positive class — a sign of the 77/12/11 class imbalance in the data
- For this business use case, minority-class recall matters more than raw accuracy — missing a real complaint is costlier than a false alarm

Handling Class Imbalance — Confusion Matrices

CONFUSION MATRICES — ROW-NORMALISED, TEST SET



EFFECT OF class_weight = "balanced"

Model	Acc.	Macro F1	Neg. rec.	Neut. rec.
Logistic Reg.	82.0%	0.57	0.41	0.23
Logistic Reg. (balanced)	77.1%	0.60	0.58	0.49
Random Forest	78.5%	0.39	0.14	0.02
Random Forest (balanced)	81.0%	0.56	0.45	0.22

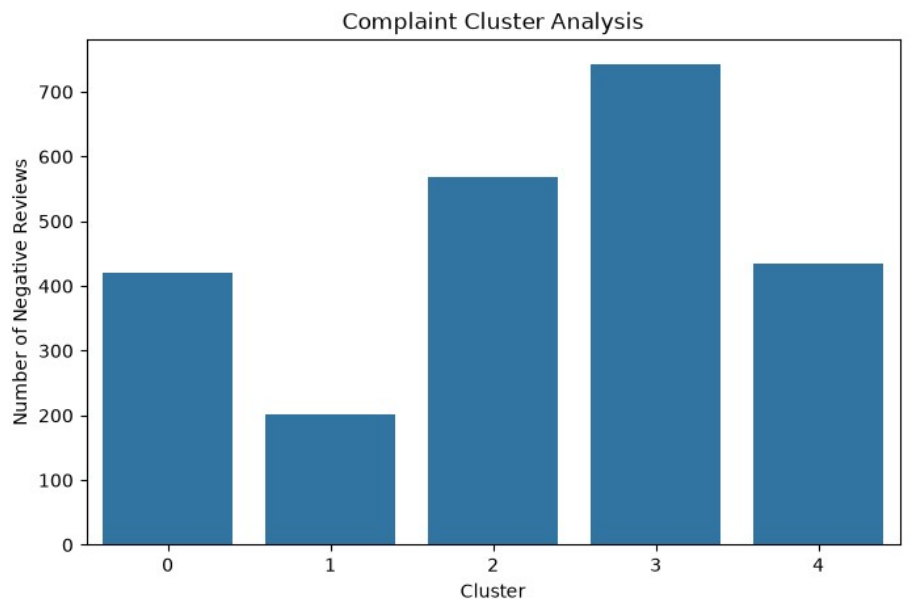
Key takeaways

- Class weights trade overall accuracy for minority-class recall: Logistic Regression Negative recall 0.41 → 0.58 and Neutral 0.23 → 0.49
- Macro F1 rises from 0.57 to 0.60 while accuracy drops from 82.0% to 77.1%
- Random Forest also gains on Negative reviews (recall 0.14 → 0.45) but still misses most Neutral ones
- Neutral (3-star) reviews are mostly predicted as Positive (61% with plain Logistic Regression)

Customer Pain Points — K-Means Clustering

- 2,369 Negative reviews isolated and re-encoded with the fitted TF-IDF vectorizer
- KMeans(n_clusters=5, random_state=42, n_init=10) groups them into 5 themes
- Per cluster, the top 10 highest-weighted TF-IDF terms define its complaint theme — sizing (small/large), fabric/material quality, boxy or short fit, and appearance not matching expectations

COMPLAINT CLUSTER ANALYSIS



TOP WORDS PER CLUSTER

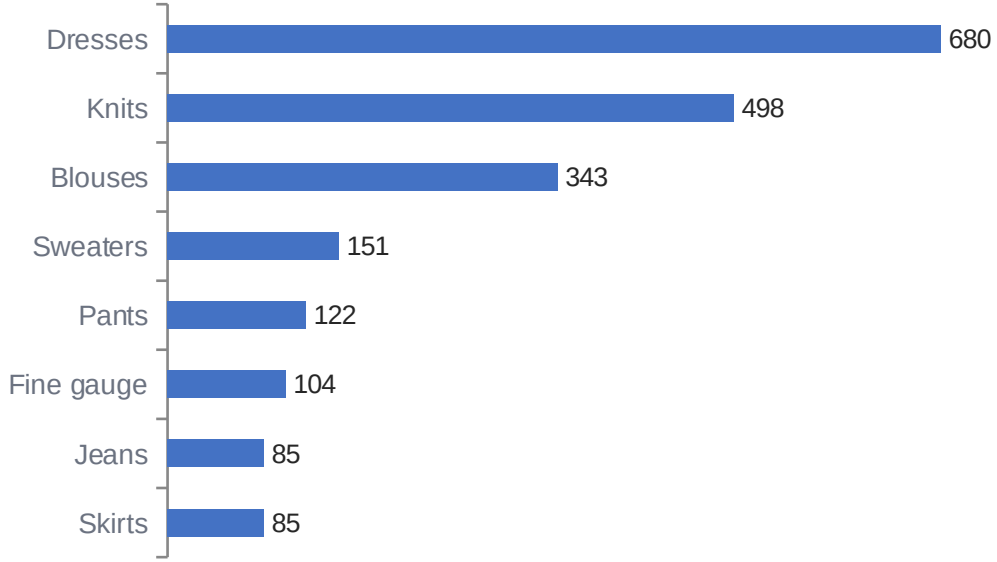
Cluster	Theme	Top TF-IDF terms
0 (420)	Look/appearance mismatch	back, like, looks, looked, going, look, fabric, model, top, cute
1 (202)	Boxy / short fit (tops)	shirt, would, back, fabric, boxy, like, short, loved, top, look
2 (569)	Sizing — runs small/large	top, small, size, would, ordered, fit, like, way, really, large
3 (743)	Fabric / material quality	quality, fabric, material, cheap, sweater, thin, price, like, even, would
4 (435)	Dress disappointment	dress, like, fabric, would, really, disappointed, look, looked, material, could

Business Insights Dashboard

WHAT WE KNOW

- Class imbalance: 77% Positive vs. 23% combined Neutral+Negative — most customers are satisfied, but the minority signal is where risk hides
- 2,369 negative reviews (10–12% of all reviews) group into 5 distinct complaint themes via clustering
- 5 interpreted themes: fabric/material quality (743 reviews, the largest), sizing small/large (569), dress disappointment (435), look/appearance mismatch (420), and boxy/short fit (202)
- Dresses (680) and Knits (498) alone account for ~50% of all negative reviews

NEGATIVE REVIEWS BY PRODUCT CATEGORY



RECOMMENDED ACTIONS FOR THE BUSINESS

- Route reviews auto-tagged Negative + their cluster theme directly to the relevant team, starting with Dresses and Knits
- Track weekly Negative-review volume per cluster and per category
- Re-run clustering monthly as new reviews arrive to catch emerging complaint themes early

Key Insights & Conclusion

- Class imbalance (77% Positive) is the main driver of weak Neutral/Negative recall
- Logistic Regression is the strongest model; class_weight='balanced' lifts Negative recall from 0.41 to 0.58 and Macro F1 to 0.60, at the cost of overall accuracy (82.0% → 77.1%)
- Automating sentiment + complaint clustering lets customer service teams react to emerging issues as they appear, instead of reading reviews manually